

Multi-locus sequence typing (MLST) of clinical isolates of methicillin-resistant *Staphylococcus aureus* (MRSA) from systemic infections in two hospitals in Quito-Ecuador

Doménica Andrade¹, Tatiana Lara², Marcelo Grijalva^{1,2}

¹Grupo de Investigación en Sanidad Humana y Animal – GISAH, Departamento de Ciencias de la Vida y la Agricultura, Universidad de las Fuerzas Armadas ESPE, Sangolquí, Ecuador.

²Centro de Nanociencia y Nanotecnología, Universidad de las Fuerzas Armadas ESPE, Sangolquí, Ecuador.

279



Summary

Methicillin-resistant *Staphylococcus aureus* (MRSA) infections are currently a serious clinical and epidemiological problem. This opportunistic pathogen has been commonly associated with hospital infections. However, community variants of high infectivity have recently appeared and have raised concerns due to their virulence. In this research, twenty-six clinical isolates from blood cultures belonging to patients from two reference hospitals in Ecuador, were analyzed using Multilocus Sequence Typing (MLST). PCR assays were standardized and optimized for the housekeeping MLST genes (*arcC*, *aroE*, *glpF*, *gmk*, *pta*, *tpiA*, and *yqiL*) and for the cyclic sequencing reaction required in the Sanger sequencing method, along with necessary purification proce-

dures. The isolates' allelic variants and their sequence types (ST) were analyzed using the PubMLST database. Eight previously undetermined sequence types (UST) were obtained, along with two alleles with point mutations. The analysis of these data using Maximum Parsimony Trees and Minimum Spanning Trees (MSTree) showed no closer phylogenetic associations for the isolates studied and the most commonly reported clones, demonstrating the emergence of new variants in Ecuador, presumably community-type ones. The present study may contribute to timely clinical and epidemiological surveillance of MRSA infections at a local level.



Key words

Staphylococcus aureus, Antibiotic resistance, Multilocus sequence typing

Corresponding author

Marcelo Grijalva, M.D., Ph.D.,
Phone: +593 984 015 558,
Email: rmgrijalva@espe.edu.ec

1. Introduction

Methicillin-resistant *Staphylococcus aureus* (MRSA) is an opportunistic Gram-positive pathogen with the ability to invade tissues and cause a wide range of infections, including serious ones such as osteomyelitis, endocarditis, necrotizing pneumonia, and bacteremia.¹⁻³ Particularly, MRSA bacteremia represents a challenge, as it can evolve into metastatic infections and life-threatening complications, with higher morbidity and mortality rates.^{4,5}

The World Health Organization,⁶ has included MRSA among high-priority microorganisms, with a 64% higher probability of death in patients carrying these strains compared to those infected by non-resistant strains. Moreover, MRSA infections are currently considered the leading cause of morbidity and mortality among antibiotic-resistant pathogens.^{7,8}

MRSA resistance to methicillin and ceftiofloxacin is conferred by the acquisition of *mecA* gene within the Staphylococcal Cassette Chromosome (SCCmec).¹ The *mecA* gene encodes an alternative penicillin-binding protein (PBP2a) which avoids the β -lactam antibiotics binding to the bacterial cell wall, thereby preventing the cell membrane collapse.^{9,10}

MRSA spreads by direct or indirect contact and has increased its presence significantly in hospitals and

communities. Hospital and community-acquired MRSA strains (HA and CA, respectively) have different types of SCCmec cassettes and express different genes that may be associated with the strain's virulent phenotype and spread success.¹¹⁻¹³ Some strains are even known to be restricted to a geographical area, while others have pandemic spread.⁷ In Ecuador, there is a limited number of studies related to HA and CA MRSA strains and even fewer related to their evolution, epidemiology, and virulence.¹⁴⁻¹⁶ Therefore, efforts are needed to address these gaps and improve diagnostic, treatment, and surveillance strategies for MRSA bacterial infections and their complications.¹⁷

Molecular typing has proven to be an important tool for monitoring the distribution, dissemination, and evolution of bacterial strains and their clones.¹⁸ In this way, the Multi-Locus Sequence Typing (MLST) technique is widely used for its unambiguous capacity for bacterial strain characterization based on internal sequencing of certain housekeeping genes. Using MLST, housekeeping gene sequences for every isolate can be assigned to defined alleles. The set of genes for each isolate generates a specific allelic profile known as Sequence Type (ST).¹⁹ The MLST technique provides easy reproducibility and interpretation, while allowing the acquisition of significant amounts of data and high discriminatory power.²⁰

The aim of the study was to characterize a collection of MRSA isolates from blood cultures of patients with bacteremia. Patients belonged to intensive care units of two tertiary care hospitals in Quito. We used the MLST typing tool with a seven-gene panel and the Sanger sequencing technique. We analyzed the isolates' evolutionary relationships and their association to specific clonal complexes.

2. Materials and methods

1. Recovery of Clinical Isolates

The collection was obtained from blood cultures of ICU patients belonging to two tertiary-care hospitals, the Eugenio Espejo (HEE) Hospital from the Ministry of Health of Ecuador and the Carlos Andrade Marín (HCAM) Hospital, the largest health facility of the Ecuadorian Institute of Social Security (IESS). The hospitals serve different populations in Quito and are referral facilities for different provinces in the country.

The initial collection included fifty-three MRSA isolates that were cryopreserved in our laboratory at Universidad de las Fuerzas Armadas ESPE. For isolates recovery, cryovials were placed at room temperature for one hour and then incubated at 36°C for 15 minutes. Each isolate was then inoculated to separate nutrient broth (Merck), nutrient agar (Merck), TSA (Beckton, Dickinson, and Company), and 5% blood agar (Beckton, Dickinson, and Company).^{21,22} The cultures were incubated at 36°C for 20 hours. Later, Gram staining was performed on those dishes with growth. The presence of Gram-positive cocci was verified using the optical microscope. Gram-positive cocci isolates were cryopreserved in BHI (Liofilchem) supplemented with 20% glycerol and stored in cryovials (CryoBank Mixed, COPAN Diagnostics Inc.) at -80°C.²³

2. Molecular Characterization

A multiplex PCR assay was applied to the recovered isolates to confirm *S. aureus* identity (*nuc* and *16S rRNA* genes) and their methicillin-resistant or methicillin-susceptible antimicrobial (MRSA or MSSA) profiles (*mecA* gene). For this purpose, DNA extraction protocols and multiplex *nuc*, *16S rRNA*, and *mecA* PCR protocols previously reported by Noboa, 2020²⁴ were used. The PCR products were visualized by electrophoresis on 2% agarose gels (Invitrogen).

3. MLST technique

2.1.1. End-point PCR

The GoTaq Flexi Polymerase kit (Promega) was used for the end-point PCR (final reaction volume of 15

μL),²⁵ following the Enright et al. procedure for 7 MLST housekeeping genes (*arcC*, *aroE*, *glpF*, *gmk*, *pta*, *tpiA*, and *yqiL*).²⁶ The thermocycler program parameters were taken from Saunders and Holmes,²⁷ and used with slight modifications. The presence of the amplified genes was verified by electrophoresis on ultra-pure agarose gels (Invitrogen) at 2%.

Amplicons were subjected to a purification process with the PureLink Quick PCR Purification Kit (Thermo Fisher Scientific).²⁸ The purified DNA was stored at 4°C or -20° for short or long-term storage, respectively.

2.1.2. Sanger sequencing

The cyclic sequencing reaction was performed with the BigDye Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems, 2016).²⁹ The forward primer was used for almost all housekeeping genes reactions, except for the *pta* and *yqiL* genes where forward and reverse sequencing was necessary. Sequencing reaction components and the program used were standardized following the manufacturer's instructions.²⁹ For the purification of cyclic PCR products, we used a Sephadex G-50 method described by Chen and Yung.³⁰

The sequencing electrophoresis was performed using the 3130 Genetic Analyzer (Applied Biosystems) with a 36cm capillary array (Applied Biosystems, CA). Raw fluorescence data were analyzed using the Sequencer Analysis Software v 5.1 (Applied Biosystems, 2003, USA).

4. Bioinformatic Analysis

2.4.1 Sequence curation

Software packages Geneious Prime v 10.1.2,³¹ and SnapGene 5.3.2 (GSL Biotech, IL) were employed to perform sequence curation and analysis.

2.4.2 Identification of allelic numbers and sequence types

The allelic numbers for each locus were obtained using the University of Oxford Pub-MLST database (available on the site https://pubmlst.org/bigdb?db=pubmlst_saureus_seqdef). Alleles obtained for each locus were downloaded and analyzed using the Geneious Prime 2020.0.4 software. The alignment with the sequences of the isolate in question was performed using the MUSCLE ALIGNMENT tool.³² The threshold to choose the best allelic number was equal to or greater than 98.2% (identity assignment). And finally, the combination of the 7 allelic numbers for each isolate was used to obtain its sequence type using the same database.

2.4.3 Phylogenetic analysis

Sequence alignments for each gene and for the con-



catenation of the 7 loci were performed with the ClustalW algorithm from the "Align" tool in MEGA-X.³³ The evolutionary history was inferred using the UPGMA method to obtain the optimal tree for each gene and for the concatenated sequences of all genes.^{34,35} Evolutionary distances were computed using the p-distance method,³⁶ and were expressed as the number of base differences per site. Central STs sequences belonging to the most frequently found clonal complexes in the PubMLST database were used for comparison through a Minimum Spanning Tree,¹⁹ which was built using the online software Phyloviz (<http://www.phyloviz.net/>).³⁷

3. Results and Discussion

Although 53 isolates were initially available, their preservation at different temperatures for extended periods had an impact on the efficiency of culture restoration. The maintenance of bacteria at temperatures below 4°C for extended periods, induces the accumulation of mutations that disrupt the cell's transcription and translation.³⁸ This is reflected in changes in the thickness of the cell wall and reduction of enzyme activity, but above all, an increase in membrane permeability caused by dehydration and cell shrinkage.³⁹

For *S. aureus* identification, we targeted the *16SrRNA* gene, which allows the identification of the minor subunit component of ribosomes in prokaryotes,⁴⁰ along with the *nuc* gene that has long been used for the identification of *Staphylococcus aureus*, as this encodes a thermonuclease characteristic of this species.²¹ Also, a region of the *mecA* gene was amplified as it encodes the PBP2a protein, which grants methicillin resistance.⁴¹ Isolates that did not amplify any gene of interest, were removed from the study. Regarding the presence of *mecA* gene, we observed no amplification for the *nuc* gene in five isolates. Hoegh et al.⁴² previously described that the absence of the *nuc* gene in these cases might be caused by partial deletions of the gene, resulting in a *nuc*-negative PCR. These isolates were also discarded for next processing. Later, we successfully amplified the seven housekeeping genes for the 26 remaining isolates that showed the MRSA ARM pattern.

Our study successfully assigned an allelic variant for each gene sequence, but no sequence type (ST) could be obtained for any isolate. The new allelic combinations were called undetermined sequences (UST). In this way, the isolates were pooled into 8 USTs with a preponderance of UST-1 and UST-4 (Table 1). It has been reported that in South American isolates, the

predominantly circulating sequence types are ST-1, ST-5, and ST-8.²⁰ In Ecuador, ST-8 (related to USA300), ST-45, ST-30, ST-5, and ST-22 have been reported along with epidemic clones such as the Brazilian (ST-239-MRSA-III), pediatric (ST-5-MRSA-IV), and the New York/Japan (ST-5-MRSA-II) ones.¹⁶ Nevertheless, other studies carried out in LATAM showed the appearance of new allele combinations. Monteiro et al.,⁴³ discovered 9 new STs in Brazil, while ST-8 predominates in Chile, but new allele combinations were also found.⁴⁴ Even though two Argentine studies found a predominance of ST-5 and ST-30, they also described several non-typeable strains.⁴⁵

The epidemiology of *S. aureus* reveals that only a few STs tend to predominate in studies worldwide.⁵ New clones and ST could appear, however, due to adaptation and also with new studies being conducted, particularly in our region, where large-scale MLST studies are still scarce.

The emergence of new allelic variant combinations involves an evolutionary process that allows the bacterium to adapt to epigenetic variables in order to survive.⁴⁶ This fact, along with the low number of isolates reported in the database from the South American area (393 in total, 1 from Ecuador), could support the appearance not only of new STs, but also of different clonal complexes and allele variation rates. Large-scale MRSA MLST analysis from different hospital and community sources will be needed to better understand the variety of STs circulating in the country.

The Table 1 shows the allelic variability among the seven genes studied. *ArcC*, *aroE*, *gmk* and *yqiL* genes presented greater variability, with 4 allelic possibilities each. The *glpF*, *pta* and *tpiA* genes had only 2 allelic variants, with the *glpF* gene being the most conserved since only one isolate had a different allele. In the research of Enright et al.,²⁶ the most diverse gene fragments corresponded to *arcC* and *aroE* and the most uniform were those of *glpF*. In the same way, the analysis carried out by Polveiro et al.,⁴⁷ demonstrated that *aroE*, *gmk* and *yqiL* were the genes with the highest diversity, showing polymorphic sites of 2.93%, 2.15%, and 2.90%, respectively, while *glpF*, *pta* and *tpiA* remained the ones with the lowest mutation rates, with 1.07%, 1.26% and 1.74%, respectively.

The allelic assignment of MLST genes revealed two isolates with one different point mutation each. The mutations were found in isolates X8 and R1, in *aroE* and *pta* genes, respectively (Figure. 1), and thus these two isolates were considered as single-locus variants (SLVs) of UST-1 with a 99.99% of similarity. Polymorphisms in metabolic genes of *S. aureus* are unexpectedly common and are generated by the adaptation of the bacterium to its environment to survive

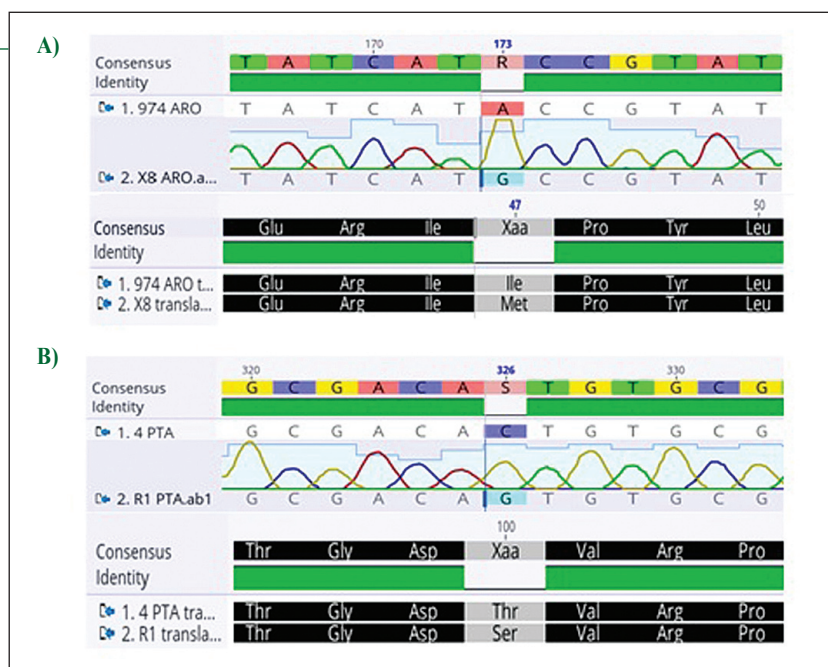
Table 1

Allelic profiles of MLST genes (*arcC*, *aroE*, *glpF*, *gmk*, *pta*, *tpiA*, and *yqiL*) obtained for the 26 isolates analyzed. ST: Sequence Type; UST: Undetermined Sequence Type.

Strain	<i>arcC</i>	<i>aroE</i>	<i>glpF</i>	<i>gmk</i>	<i>pta</i>	<i>tpiA</i>	<i>yqiL</i>	ST
A11	892	974	933	78	4	819	989	UST-1
A17	892	974	933	78	4	819	989	UST-1
B5	892	974	933	78	4	819	989	UST-1
B7	892	974	933	78	4	819	989	UST-1
B8	892	974	933	78	4	819	989	UST-1
B9	892	974	933	78	4	819	989	UST-1
R5	892	974	933	78	4	819	989	UST-1
R9	892	974	933	78	4	819	989	UST-1
R10	838	974	933	78	4	819	989	UST-1
X9	892	974	933	78	4	819	989	UST-1
X10	892	974	933	78	4	819	989	UST-1
R1	892	974	933	78	New allele : 99% similarit y with allele4	819	989	SLV-1 of UST-1
X8	892	New allele : 99% similarit y with allele 974	933	78	4	819	989	SLV -2 of UST-1
A19	620	974	933	78	4	819	989	UST-2
A13	620	974	933	542	4	819	989	UST-3
A12	838	377	933	8	899	815	992	UST-4
A14	838	377	933	8	899	815	992	UST-4
A15	838	377	933	8	899	815	992	UST-4
B6	838	377	933	8	899	815	992	UST-4
B11	838	377	933	8	899	815	992	UST-4
R2	838	377	933	8	899	815	992	UST-4
R8	838	377	933	8	899	815	992	UST-4
A16	838	971	933	8	4	819	989	UST-5
A18	838	971	933	8	899	815	992	UST-6
R6	22	716	14	470	899	819	747	UST-7
B10	838	971	933	8	899	815	483	UST-8

Figure 1

Point mutations observed in the study and their translation products. A) Allelic variant with point mutation A→G at nucleotide 173 of the *aroE* gene in the X8 sample, non-synonymous mutation. B) Allelic variant with point mutation C→G at nucleotide 326 of the *pta* gene in sample R1, non-synonymous mutation.



stress, maintain its pathogenicity or adapt its growth. Furthermore, considering that the stored samples underwent several procedures from the sampling to the cryopreservation and the time elapsed from sampling to genotyping, point mutations in the bacterial genetic material might be expected.^{48,49}

Mutations were confirmed by repeating both amplification and sequencing, including forward and reverse primers for each gene. Each mutation was translated to determine if it was synonymous or non-synonymous. Translated mutations present a variation in the encoded amino acid, which gives rise to altered proteins. Despite that, the BLAST analysis of the *aroE* gene mutation yielded 99.77% similarity to the identified shikimate dehydrogenase protein. In the case of the mutation found in the *pta* gene, a 99.97% similarity with the acetyltransferase phosphate protein was obtained.

As expected in each case, the alignments were analogous to the relevant metabolic proteins. It is important to consider the potential implications these mutations could have. They could affect the survival rate and the rate of extracellular acidification during host infection, also affecting the ability of the bacteria to form biofilms.^{50,51}

The evolutionary relationships of each gene are evidenced in individual phylogenetic trees (Figure. 2, A-F), where isolates are grouped based on the alleles that were assigned by the database. The same analysis was conducted for the set of isolates using the concatenated MLST sequences (Figure. 2, G), where a

range of taxa was observed, suggesting the existence of many independent evolutionary events arising within the evolutionary history of each strain.⁵² It is also remarkable that isolates provided by the same facility, are not related to each other in our trees, strengthening the argument that the isolates studied belong to CA-MRSA and are evolving individually. In their study, Florida et al.,⁵³ conclude that intraspecific evolutionary stories can be modeled by the isolate's lifestyle, which conditions their horizontal gene transfer possibilities, genome alterations, and selection of accessory genes. In turn, the pattern of bifurcations leads to the emergence of different taxa, grouped based on the USTs identified for each isolate. Although in this work the selected isolates could not be cataloged within specific clonal groups, cluster formation can be observed (Figure 2).

In the Minimum Spanning Tree (Figure.3), no close clonal relationship between our study isolates and the previously reported STs was found. Furthermore, the values of divergency presented in Table 2, show that ST-5 and ST-8 are the closest STs to the study group, USTs from 1 to 3, including the samples with mutations, are closely related to ST-8, while USTs from 4-8 have better correlation with ST-5. Interestingly, most of the isolates could not be grouped into specific clonal complexes (CC). Arévalo,⁵⁴ suggests that the divergence within isolates may be related to a spread of hospital strains (HA-MRSA) at the community level (CA-MRSA), where they acquire new genetic characteristics, including greater genetic variability. An increased prev-

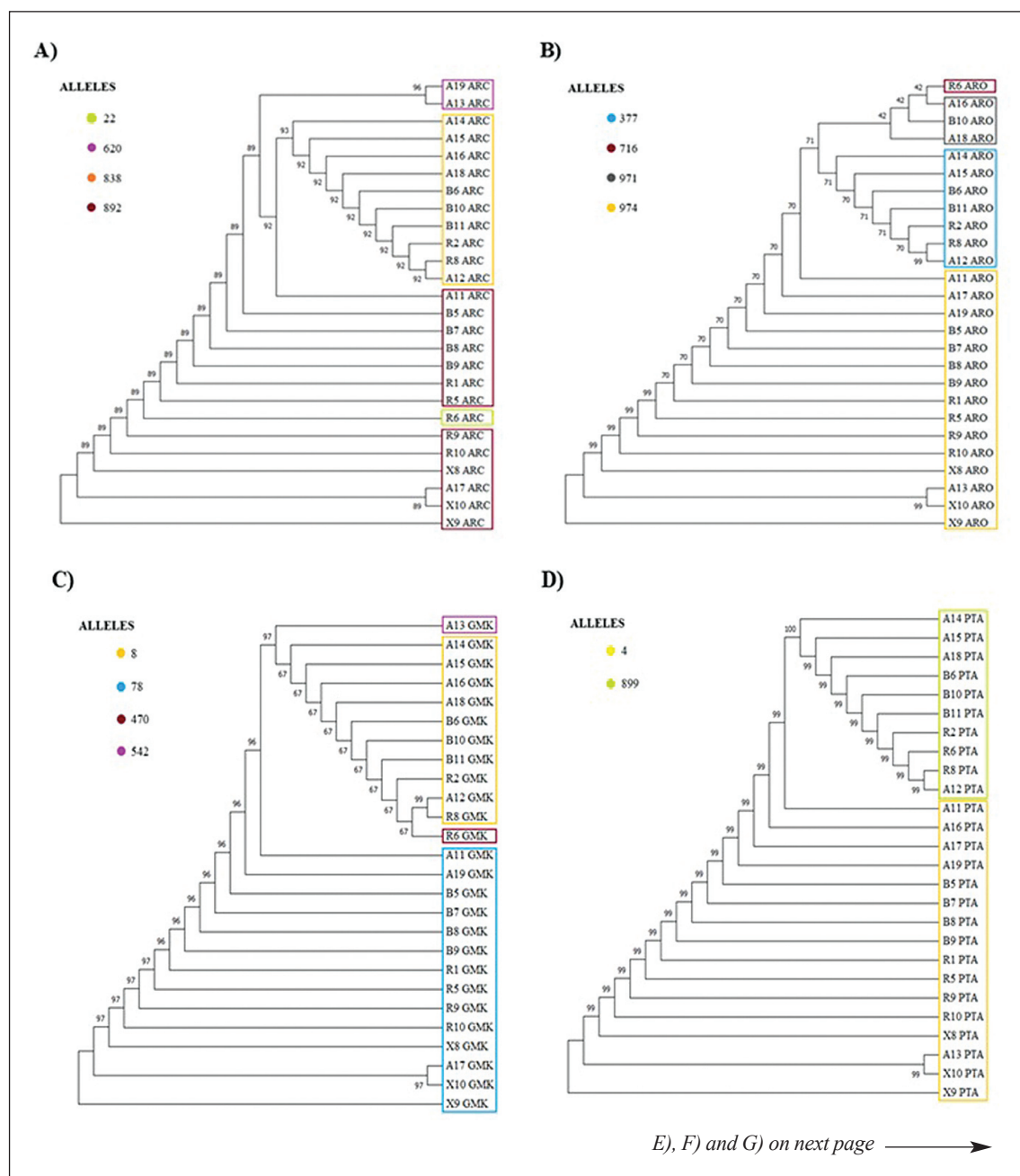


Figure 2 Maximum Parsimony Trees obtained for A) *arcC* gene; B) *aroE* gene; C) *gmK* gene; D) *pta* gene; E) *tpiA* gene; F) *yqiL* gene and G) concatenated sequences of the 26 isolates. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (500 replicates) are shown next to the branches.

absence of CA-MRSA could generate a greater number of potentially multi-resistant and virulent variants. Data from studies in Brazil, Ecuador, Venezuela, and Colombia rate the CA-MRSA isolates at 27%.⁵⁵

One of the most notable characteristics of MRSA clones in recent decades has been the spread of community strains over hospital strains, as well as the acquisition of new resistant genes and virulence factors

to improve their regulatory systems.⁵⁶ Lakhundi and Zhang agreed that HA-MRSA is primarily distributed in two clonal complexes around the world, whereas CA-MRSA is more dispersed between CC and has only a few genetic similarities. They also stated that due to the rapid evolution of MRSA adaptation, new strains may emerge. Furthermore, new clonal complexes may emerge as a result of genetic variation and recombina-

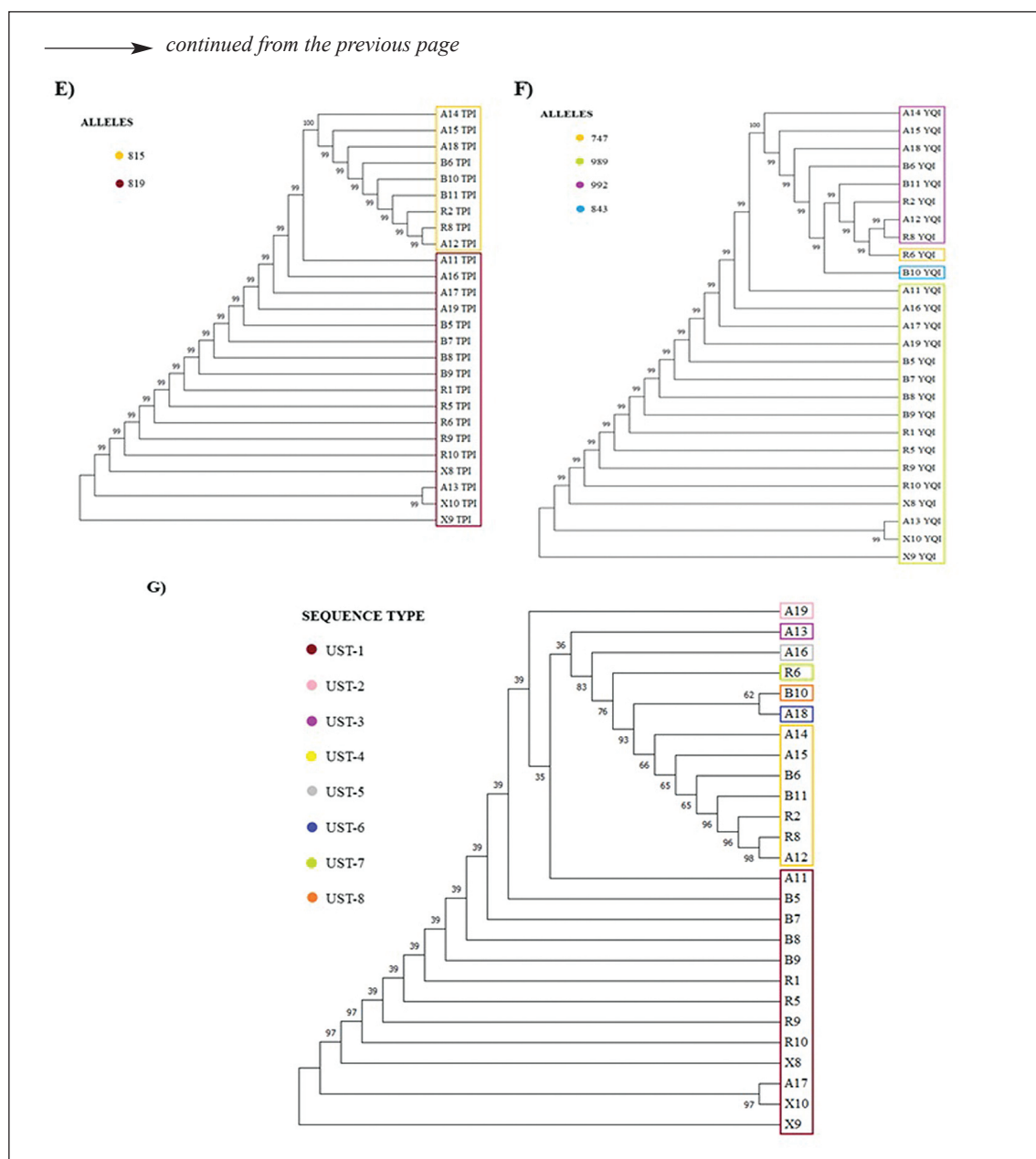


Figure 2 Maximum Parsimony Trees obtained for A) *arcC* gene; B) *aroE* gene; C) *gmK* gene; D) *pta* gene; E) *tpiA* gene; F) *yqiL* gene and G) concatenated sequences of the 26 isolates. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (500 replicates) are shown next to the branches.

nation combined with strain demographic pressure, which can promote the replacement of successful strains with emerging new ones (a phenomenon called clonal replacement).⁵⁷

4. Conclusions

Molecular typing of MRSA isolates determined the

presence of eight new sequence types. No defined clonal complexes were found, nor was a significant relationship with previously described clones in the region demonstrated. These findings, as well as the variability of strain clonal complexes, could be attributed to CA-MRSA origin.

The existence of new allelic variants was evidenced by the identification of non-synonymous mutations in the *aroE* and *pta* gene alleles. New variants, as well

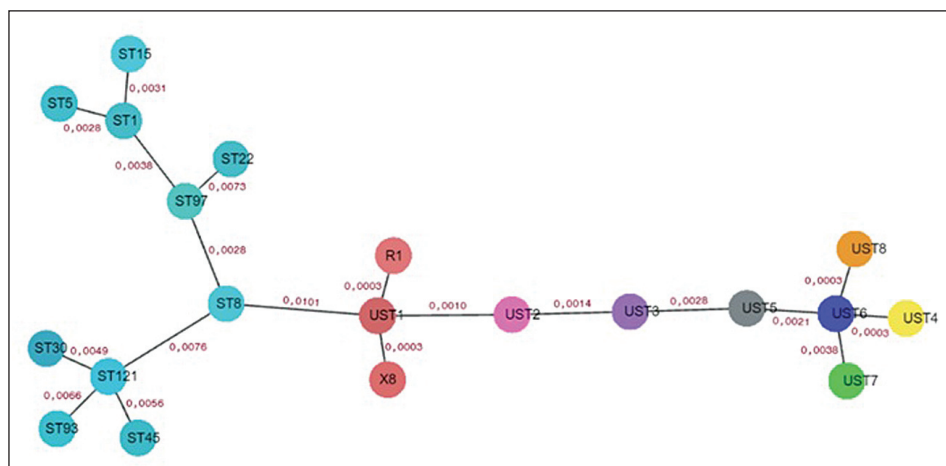


Figure 3 Minimum Spanning Tree (MSTree) generated by goeBURST A) MSTree of 26 MRSA isolates of the present study; B) MST of 26 MRSA isolates of the present study compared to clonal complexes central clone.

Table 2

Estimates of Evolutionary Divergence between Sequences. The closest divergences between the isolates in this study and the central STs are highlighted.

	ST1	ST5	ST8	ST15	ST22	ST30	ST45	ST93	ST97	ST121	UST1	UST2	UST3	UST4	UST5	UST6	UST8	UST7	R1	X8
ST1	0,0000																			
ST5	0,0028	0,0000																		
ST8	0,0045	0,0052	0,0000																	
ST15	0,0031	0,0045	0,0056	0,0000																
ST22	0,0083	0,0076	0,0087	0,0094	0,0000															
ST30	0,0101	0,0094	0,0083	0,0104	0,0101	0,0000														
ST45	0,0087	0,0080	0,0090	0,0090	0,0094	0,0056	0,0000													
ST93	0,0128	0,0115	0,0104	0,0132	0,0135	0,0090	0,0094	0,0000												
ST97	0,0038	0,0066	0,0028	0,0062	0,0073	0,0090	0,0097	0,0118	0,0000											
ST121	0,0094	0,0087	0,0076	0,0097	0,0094	0,0049	0,0056	0,0066	0,0083	0,0000										
UST1	0,0146	0,0153	0,0101	0,0156	0,0187	0,0184	0,0191	0,0198	0,0128	0,0177	0,0000									
UST2	0,0156	0,0163	0,0111	0,0167	0,0198	0,0174	0,0194	0,0208	0,0139	0,0187	0,0010	0,0000								
UST3	0,0170	0,0149	0,0125	0,0180	0,0198	0,0167	0,0187	0,0201	0,0153	0,0180	0,0024	0,0014	0,0000							
UST4	0,0132	0,0104	0,0156	0,0149	0,0180	0,0198	0,0184	0,0212	0,0170	0,0191	0,0056	0,0066	0,0052	0,0000						
UST5	0,0149	0,0121	0,0132	0,0160	0,0170	0,0187	0,0174	0,0201	0,0153	0,0180	0,0031	0,0042	0,0028	0,0024	0,0000					
UST6	0,0128	0,0101	0,0153	0,0146	0,0177	0,0194	0,0180	0,0208	0,0167	0,0187	0,0052	0,0062	0,0049	0,0003	0,0021	0,0000				
UST8	0,0132	0,0104	0,0156	0,0149	0,0180	0,0198	0,0184	0,0212	0,0170	0,0191	0,0056	0,0066	0,0052	0,0007	0,0024	0,0003	0,0000			
UST7	0,0146	0,0139	0,0149	0,0170	0,0187	0,0191	0,0198	0,0212	0,0149	0,0184	0,0049	0,0059	0,0052	0,0042	0,0045	0,0038	0,0042	0,0000		
R1	0,0149	0,0156	0,0104	0,0160	0,0191	0,0187	0,0194	0,0201	0,0132	0,0180	0,0003	0,0014	0,0028	0,0059	0,0035	0,0056	0,0059	0,0052	0,0000	
X8	0,0149	0,0156	0,0104	0,0160	0,0191	0,0187	0,0194	0,0201	0,0132	0,0180	0,0003	0,0014	0,0028	0,0059	0,0035	0,0056	0,0059	0,0052	0,0007	0,0000

as new sequence types found in this study, should be reported to the PubMLST database.

The present study may contribute to timely clinical and epidemiological surveillance of MRSA infections at a local level. Nevertheless, it has limitations in terms of the number of isolates, and new larger-scale NGS-based investigations should follow as they can provide larger amplicon sizes. This might result in fewer differences in sequence coverage among alleles and an increased likelihood of higher numbers of mutations identified within each gene.

Acknowledgments: We would like to thank Francisco Flores, Judith Torres, Dayana Aguilar, and Kirsty Noboa for their helpful comments and guidance during the realization of this study.

Declaration of conflict of interest: The authors have no conflicts of interest to declare.

Author agreement: All authors have seen and approved the final version of the manuscript being submitted. The manuscript is an original work that has not been published before, and is not under consideration for publication elsewhere.



Περίληψη

Multi-locus sequence typing (MLST) of clinical isolates of methicillin-resistant *Staphylococcus aureus* (MRSA) from systemic infections in two hospitals in Quito-Ecuador

Doménica Andrade¹, Tatiana Lara², Marcelo Grijalva*¹⁻²

¹Grupo de Investigación en Sanidad Humana y Animal – GISAH, Departamento de Ciencias de la Vida y la Agricultura, Universidad de las Fuerzas Armadas ESPE, Sangolquí, Ecuador.

²Centro de Nanociencia y Nanotecnología, Universidad de las Fuerzas Armadas ESPE, Sangolquí, Ecuador.

*Υπεύθυνος αλληλογραφίας

Οι λοιμώξεις από *Staphylococcus aureus* ανθεκτικούς στη μεθικιλίνη (MRSA) αποτελούν σήμερα ένα σοβαρό κλινικό και επιδημιολογικό πρόβλημα. Αυτό το ευκαιριακό παθογόνο εντοπίζεται συνήθως με νοσοκομειακές λοιμώξεις. Ωστόσο, πρόσφατα εμφανίστηκαν παραλλαγές στην κοινότητα υψηλής μολυσματικότητας και έχουν εγείρει ανησυχίες λόγω της λοιμογόνου δυναμικής τους. Στην παρούσα μελέτη, είκοσι έξι κλινικά απομονωμένα στελέχη από καλλιέργειες αίματος που ανήκαν σε ασθενείς από δύο νοσοκομεία αναφοράς στον Ισημερινό, αναλύθηκαν με τη χρήση Multilocus Sequence Typing (MLST). Οι μοριακές αναλύσεις με PCR τυποποιήθηκαν και βελτιστοποιήθηκαν για τα γονίδια MLST (*arcC*, *aroE*, *glpF*, *gmk*, *pta*, *tpiA*, *yqiL*) και για την αντίδραση κυκλικής αλληλουχίας που απαιτείται στη μέθοδο προσδιορισμού αλληλουχίας κατά Sanger, μαζί με τις απαραίτητες διαδικασίες καθαρισμού. Οι αλληλόμορφες παραλλαγές, καθώς και οι τύποι αλληλουχίας τους (sequence types, ST) αναλύθηκαν χρησιμοποιώντας τη βάση δεδομένων PubMLST. Απομονώθηκαν οκτώ προηγούμενως μη καθορισμένοι τύποι αλληλουχίας (undetermined ST, UST), μαζί με δύο αλληλόμορφα με σημειακές μεταλλάξεις. Η ανάλυση αυτών των δεδομένων χρησιμοποιώντας Maximum Parsimony Trees και Minimum Spanning Trees (MSTree) δεν έδειξε στενές φυλογενετικές συσχετίσεις για τα στελέχη που μελετήθηκαν και τους πιο συχνά αναφερόμενους κλώνους, αποδεικνύοντας την εμφάνιση νέων παραλλαγών στον Εκουαδόρ, πιθανώς τύπου της κοινότητας. Η παρούσα μελέτη μπορεί να συμβάλει στην έγκαιρη κλινική και επιδημιολογική επιτήρηση των λοιμώξεων από MRSA σε τοπικό επίπεδο.



Λέξεις κλειδιά

Staphylococcus aureus, Αντοχή στα αντιβιοτικά, Τυποποίηση αλληλουχίας πολλαπλών τόπων

References

1. Algamal AM, Hetta HF, Elkelish A, Alkhalifah DHH, Hozzein WN, Batiha GES, *et al.* Methicillin-resistant *Staphylococcus aureus* (MRSA): One health perspective approach to the bacterium epidemiology, virulence factors, antibiotic-resistance, and zoonotic impact. *Infect Drug Resist* 2020; 13:3255–3265.
2. Cervantes E, García R, Salazar P. Importancia de *Staphylococcus aureus* meticilina resistente intrahospitalario y adquirido en la comunidad. *Rev Mex Patol Clin Med Lab* 2014; 61:196-204.
3. Silva V, Hermenegildo S, Ferreira C, Manaia CM, Capita R, Alonso C, *et al.* Genetic characterization of methicillin-resistant *Staphylococcus aureus* isolates from human bloodstream infections: Detection of MLSB resistance. *Antibiotics* 2020; 9:1–9.
4. Kimmig A, Hagel S, Weis S, Bahrs C, Löffler B, Pletz MW. Management of *Staphylococcus aureus* Bloodstream Infections. *Front Med (Lausanne)* 2021; 7:616524.
5. Turner NA, Sharma-Kuinkel BK, Maskarinec SA, Eichenberger EM, Shah PP, Carugati M, *et al.* Methicillin-resistant *Staphylococcus aureus*: an overview of basic and clinical research. *Nat Rev Microbiol* 2019; 17:203–218.
6. WHO: World Health Organization [Internet]. Geneva: World Health Organization; 2017. Press release, WHO publishes list of bacteria for which new antibiotics are urgently needed; 2017 Feb 27[cited 2022 Jun 8]; [about 3 screens]. Available from: <https://www.who.int/news/item/27-02-2017-who-publishes-list-of-bacteria-for-which-new-antibiotics-are-urgently-needed>
7. Goudarzi M, Seyedjavadi SS, Nasiri MJ, Goudarzi H, Sajadi Nia R, Dabiri H. Molecular characteristics of methicillin-resistant *Staphylococcus aureus* (MRSA) strains isolated from patients with bacteremia based on MLST, SCCmec, spa, and agr locus types analysis. *Microb Pathog* 2017; 104:328–335.
8. Tsouklidis N, Kumar R, Heindl SE, Soni R, Khan S. Understanding the Fight Against Resistance: Hospital-Acquired Methicillin-Resistant *Staphylococcus aureus* vs. Community-Acquired Methicillin-Resistant *Staphylococcus aureus*. *Cureus* 2020; 12:e8867.
9. Garbacz K, Wierzbowska M, Kwapisz E, Kosecka-Strojek M, Bronk M, Saki M, *et al.* Distribution and antibiotic-resistance of different *Staphylococcus* species identified by matrix assisted laser desorption ionization-time of flight mass spectrometry (MALDI-TOF MS) isolated from the oral cavity. *J Oral Microbiol* 2021; 13:1983322.
10. Kolesnik-Goldmann N, Bodendoerfer E, Röthlin K, Herren S, Imkamp F, Marchesi M, *et al.* Rapid Detection of PBP2a in *Staphylococci* from Shortly Incubated Subcultures of Positive Blood Cultures by an Immunochromatographic Assay. *Microbiol Spectr* 2021; 9:e0046221.
11. Sit P, Teh C, Idris N, Sam IC, Syed S, Sulaiman H, *et al.* Prevalence of methicillin-resistant *Staphylococcus aureus* (MRSA) infection and the molecular characteristics of MRSA bacteraemia over a two-year period in a tertiary teaching hospital in Malaysia. *BMC Infect Dis* 2017; 17:274.
12. Vivas M, Gutierrez A. Typification Methods and Molecular Epidemiology of *Staphylococcus aureus* with Methicillin Resistance. In: Hemeg H, Ozbak H, Afrin F, editors. *Staphylococcus aureus* [Internet]. London: IntechOpen; 2018 [cited 2022 Jun 6]. Available from: <https://www.intechopen.com/chapters/60849>.doi: 10.5772/intechopen.76442
13. Recker M, Laabei M, Toleman MS, Reuter S, Sanderson RB, Blane B, *et al.* Clonal differences in *Staphylococcus aureus* bacteraemia-associated mortality. *Nat Microbiol* 2017; 2:1381–1388.
14. Vaca S, Cruz S, Iñiguez Samuel. Prevalencia de *Staphylococcus aureus* meticilino resistente en el personal de salud de un Hospital de Especialidades en Quito-Ecuador. *Revista San Gregorio* 2021; 1:86–98.
15. Reyes J, Rincón S, Díaz L, Panesso D, Contreras GA, Zurita J, *et al.* Dissemination of methicillin-resistant *Staphylococcus aureus* USA300 sequence type 8 lineage in Latin America. *Clin Infect Dis* 2009; 49:1861–1867.
16. Zurita J, Barba P, Ortega D, Mora M, Rivadeneira S. Local circulating clones of *Staphylococcus aureus* in Ecuador. *Braz J Infect Dis* 2016; 20:525–533.
17. Asadollahi P, Farahani NN, Mirzaii M, Khoramrooz SS, van Belkum A, Asadollahi K, *et al.* Distribution of the Most Prevalent Spa Types among Clinical Isolates of Methicillin-Resistant and -Susceptible *Staphylococcus aureus* around the World: A Review. *Front Microbiol* 2018; 9:163.
18. Arcenas RC. Molecular Methods for Healthcare-Acquired Infections. In: Coleman W, Tsongalis G, editors. *Diagnostic Molecular Pathology*. US: Elsevier; 2017. p. 163–177.
19. Jolley KA, Bray JE, Maiden MCJ. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. *Wellcome Open Res* 2018; 3:124.



20. Rodríguez F, Fernández S, Haim S, Mollerach M, Basualdo W, Castro H, *et al.* Estandarización del análisis multi-locus de número variable de repeticiones en tándem para el estudio de *Staphylococcus aureus* resistentes a meticilina aislados de la comunidad en Paraguay. *Duazary* 2017; 14:1-10.
21. Public Health England, National Health Service. Identification of *Staphylococcus* species, *Micrococcus* species and *Rothia* species [Internet]. UK Standards for Microbiology Investigations. 2019 [cited 2022 Feb 15]. Available from: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/832968/ID_7_dj_.pdf
22. Vitko NP, Richardson AR. Laboratory Maintenance of Methicillin-Resistant *Staphylococcus aureus* (MRSA). *Curr Protoc Microbiol.* 2013; Chapter 9: Unit-9 C.2.
23. Missiakas DM, Schneewind O. Growth and Laboratory Maintenance of *Staphylococcus aureus*. *Curr Protoc Microbiol.* 2013; Chapter 9:Unit-9C.1.
24. Noboa K. Estudio de expresión génica de factores de virulencia de aislados clínicos de *Staphylococcus aureus* causantes de bacteremia. Sangolquí: Universidad de las Fuerzas Armadas ESPE; 2020. 50-120 p. <http://repositorio.espe.edu.ec/xmlui/bitstream/handle/21000/22528/T-ESPE-043854.pdf?sequence=1&isAllowed=y>
25. Promega. GoTaqFlexi DNA Polymerase (M829) Protocol [Internet]. 2018 [cited 2021 Dec 15]. Available from: https://worldwide.promega.com/-/media/files/resources/protocols/product-information-sheets/g/gotaq-flexi-dna-polymerase-m829.pdf?rev=f4d69130c5f7402898bc2116cc0e3584&sc_lang=en
26. Enright MC, Day NP, Davies CE, Peacock SJ, Spratt BG. Multilocus sequence typing for characterization of methicillin-resistant and methicillin-susceptible clones of *Staphylococcus aureus*. *J Clin Microbiol* 2000; 38:1008–15.
27. Saunders NA, Holmes A. Multilocus sequence typing (MLST) of *Staphylococcus aureus*. *Methods Mol Biol* 2007; 391:71-85.
28. Life Technologies Corporation. PureLink® PCR Purification Kit For rapid, efficient purification of PCR products Catalog numbers K3100-01 and K3100-02 [Internet]. 2011 [cited 2021 Dec 15]. Available from: https://assets.fishersci.com/TFS-Assets/LSG/manuals/purelink_pcr_man.pdf?_ga=2.130614344.690066872.1639688301-1651989391.1639688301
29. Applied Biosystems. Quick Reference: BigDye™ Terminator v3.1 Cycle Sequencing Kit [Internet]. 2016 [cited 2021 Dec 15]. Available from: https://assets.fishersci.com/TFS-Assets/LSG/manuals/MAN0015666_BigDyeTerminator_V3_1_CycleSeq_QR.pdf
30. Chen N, Yun L. DNA recovery from agarose gels with a simple centrifuge-driven Sephadex filtration. *For Stud China* 2006; 8:32–34.
31. Kears M, Moir R, Wilson A, Stones-Havas S, Cheung M, Sturrock S, *et al.* Geneious Basic: An integrated and extendable desktop software platform for the organization and analysis of sequence data. *Bioinformatics* 2012; 28:1647–1649.
32. Edgar RC. MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res* 2004; 32:1792–1797.
33. Larkin MA, Blackshields G, Brown NP, Chenna R, McGettigan PA, McWilliam H, *et al.* Clustal W and Clustal X version 2.0. *Bioinformatics* 2007; 23:2947–2948.
34. Kannan L, Wheeler WC. Maximum Parsimony on Phylogenetic networks. *Algorithms Mol Biol* 2012; 7:9.
35. Sneath P, Sokal R. Numerical Taxonomy. The Principles and Practice of Numerical Classification. 2nd ed. San Francisco:Freeman; 1973.
36. Nei M, Kumar S. Molecular Evolution and Phylogenetics. Oxford: Oxford University Press; 2000.
37. Francisco A, Bugalho M, Ramirez M, Carriço JA. Global optimal eBURST analysis of multilocus typing data using a graphic matroid approach. *BMC Bioinformatics* 2009; 10:152.
38. Guo N, Wei Q, Xu Y. Optimization of cryopreservation of pathogenic microbial strains. *Int J Biosaf Biosecurity* 2020; 2:66–70.
39. Suo B, Yang H, Wang Y, Lv H, Li Z, Xu C, *et al.* Comparative Proteomic and Morphological Change Analyses of *Staphylococcus aureus* During Resuscitation From Prolonged Freezing. *Front Microbiol* 2018; 9:866.
40. Sune D, Rydberg H, Augustinsson ÅN, Serrander L, Jungeström MB. Optimization of 16S rRNA gene analysis for use in the diagnostic clinical microbiology service. *J Microbiol Methods* 2020; 170:105854.
41. Preeja PP, Kumar SH, Shetty V. Prevalence and Characterization of Methicillin-Resistant *Staphylococcus aureus* from Community- and Hospital-Associated Infections: A Tertiary Care Center Study. *Antibiotics* 2021; 10:197.
42. Hoegh S., Skov MN, Boye K, Worning P, Jensen TG, Kemp M. Variations in the *Staphylococcus aureus*-specific nuc gene can potentially lead to misidentification of methicillin-susceptible and -resistant *S. aureus*. *Int J Med Micr*:1020–1022.
43. Monteiro A, Pinto B, Monteiro J, Ferreira R, Ribeiro P, Bando S, *et al.* Phylogenetic and Molecular Profile of *Staphylococcus aureus* Isolated from Bloodstream Infections in Northeast Brazil. *Microorganisms* 2019; 7:210.

44. Instituto de Salud Pública de Chile. Vigilancia de *Staphylococcus aureus* meticilina resistente adquirido en la comunidad. Report No.: 10. [Internet]. Chile: Instituto de Salud Pública de Chile; 2017 [cited 2022 Mar 6]. Available from: [https://www.ispch.cl/sites/default/files/BoletinStahylococcusResistente-20062018A%20\(1\).pdf](https://www.ispch.cl/sites/default/files/BoletinStahylococcusResistente-20062018A%20(1).pdf)
45. Tomatis C, Baroni M, Mendosa M, Nagel A, Mollerach A, Alvarez C, *et al.* Tipos de spa no reportados en nuestro país en *Staphylococcus aureus* de pacientes adultos de un hospital escuela, Santa Fe, Argentina. *Rev Argent Microbiol* 2018; 50:244–248.
46. Bobay L, Ochman H. Impact of Recombination on the Base Composition of Bacteria and Archaea. *Mol Biol Evol* 2017; 34:2627–2636.
47. Polveiro RC, Granja MMC, Roldão TCB, da Silva Lopes I, Vidigal PMP, Lima MC, *et al.* Multilocus sequence analysis reveals genetic diversity in *Staphylococcus aureus* isolate of goat with mastitis persistent after treatment with enrofloxacin. *Sci Rep* 2021; 11:17252.
48. Acker KP, Wong Fok Lung T, West E, Craft J, Narechania A, Smith H, *et al.* Strains of *Staphylococcus aureus* that Colonize and Infect Skin Harbor Mutations in Metabolic Genes. *iScience* 2019; 19:281–290.
49. Salazar MJ, Machado H, Dillon NA, Tsunemoto H, Szubin R, Dahesh S, *et al.* Genetic Determinants Enabling Medium-Dependent Adaptation to Nafcillin in Methicillin-Resistant *Staphylococcus aureus*. *mSystems* 2020; 5:e00828-19.
50. Gabryszewski S, Wong T, Annavajhala M, Tomlinson K, Riquelme S, Khan I, *et al.* Metabolic Adaptation in Methicillin-Resistant *Staphylococcus aureus* Pneumonia. *Am J Respir Cell Mol Biol* 2019; 61:185–197.
51. Kim G, Hooven TA, Norambuena J, Li B, Boyd J, Yang J, *et al.* Growth and Stress Tolerance Comprise Independent Metabolic Strategies Critical for *Staphylococcus aureus* Infection. *mBio* 2021; 12:e0081421.
52. Khan N, Ahsan M, Yoshizawa S, Hosoya S, Yokota A, Kogure K. Multilocus Sequence Typing and Phylogenetic Analyses of *Pseudomonas aeruginosa* Isolates from the Ocean. *Appl Environ Microbiol* 2008; 74:6194–6205.
53. Floridia-Yapur N, Rusman F, Diosque P, Tomasini N. Genome data vs MLST for exploring intraspecific evolutionary history in bacteria: Much is not always better. *Infect Genet Evol* 2021; 93:104990.
54. Arévalo J. Genotipificación de *Klebsiella Pneumoniae* mediante el análisis de secuencias de Locus Múltiples (MLST) en una colección de aislados clínicos provenientes de infecciones sistémicas de dos hospitales de referencia del Ecuador. [Sangolquí]: Universidad de las Fuerzas Armada ESPE; 2018. <http://repositorio.espe.edu.ec/handle/21000/14116>
55. Cabrejos L, Vives C, Inga J, Astocondor L, Hinostroza N, García C. Frecuencia de *Staphylococcus aureus* meticilinoresistente adquirido en la comunidad en un hospital de tercer nivel en Perú. *Rev Peru Med Exp Salud Pública* 2021; 38:313–317.
56. Rodríguez E, Seas C, Guzmán-Blanco M, Mejía C, Alvarez C, Bavestrello L, *et al.* Evolution of methicillin-resistant *Staphylococcus aureus* clones in Latin America. *Int J Infect Dis* 2010; 14:e560–e566.

